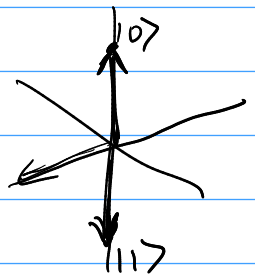
Single Spin Qubits (1e⁻, 1dot)



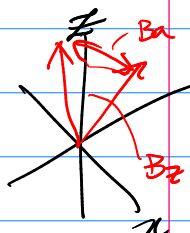
$\hat{\phi}$ or ϕ

$$H = B_x \sigma_x + B_z \sigma_z$$



Coil. L is very large.
⇒ can't off/on quickly.

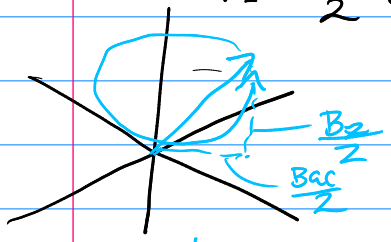
$$H = B_z \sigma_z + B_{ac} \cdot \cos(\omega t) \cdot \sigma_x$$



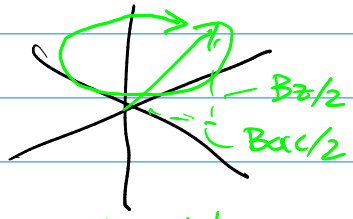
$$= B_z \sigma_z + \frac{B_{ac}}{2} \cdot \left(\cos(\omega t)\sigma_x + \sin(\omega t)\sigma_y + \cos(\omega t)\sigma_x - \sin(\omega t)\sigma_y \right)$$

$H = H_+ + H_-$

$$H_+ = \frac{B_z}{2}\sigma_z + \frac{B_{ac}}{2} \cdot \left(\cos(\omega t)\sigma_x \pm \sin(\omega t)\sigma_y \right)$$



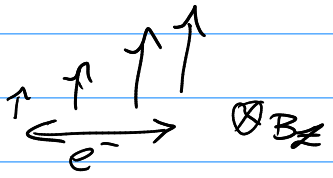
Co-rotating



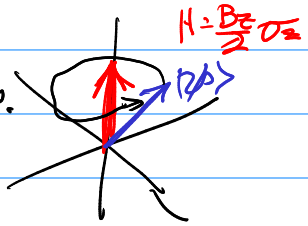
counter-rotating.



EDSR



What if we rotated with the precessing qubit?
Then it will slow/speed up i.e. / cf. Galilean Relativity.



ω of rotating frame.

$$H_{rot} = \frac{B_z - \hbar\omega}{2}\sigma_z \qquad \text{co-rotating frame}$$

$$\left(H_{rot} = \frac{B_z + \hbar\omega}{2}\sigma_z \qquad \text{counter-rotating frame} \right)$$

For drive term; in the co-rotating frame, it appears static - i.e. / a fixed rotation axis at $B_{ac}/2\ \sigma_x$.

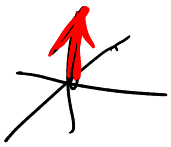
$$\left[H_{rot} = \frac{B_{ac}}{2}\sigma_x + \frac{B_z - \hbar\omega}{2}\sigma_z \right]$$

$\frac{B_z - \hbar\omega}{2} \to 0$

if $\hbar\omega = B_z$ ⇒ pure x-rotations (resonance)

In counter rotating frame...

$$H_{rot} = \frac{B_{ac}}{2}\sigma_x + \frac{B_z + \hbar\omega}{2}\sigma_z$$

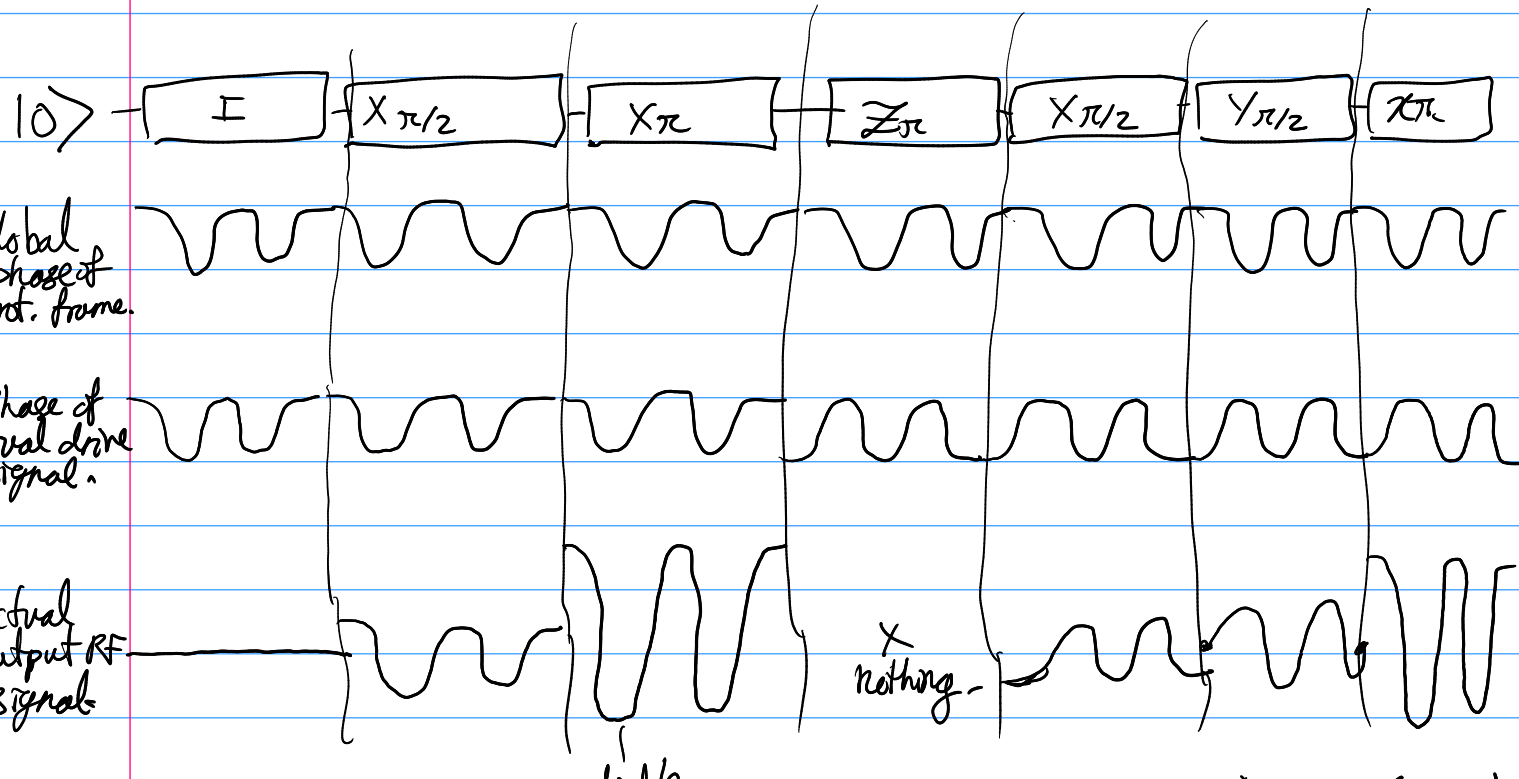


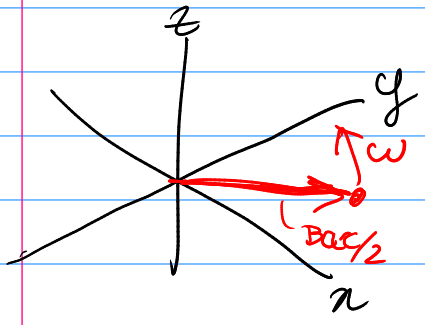
$B_z + \hbar\omega \gg B_{ac}$

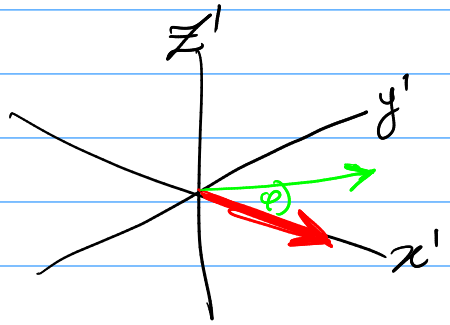
'weak driving regime'

rotating wave approximation → suppresses counter-rot. terms.

$$H_{eff} = \frac{B_{ac}}{2}\sigma_x + \frac{B_z - \hbar\omega}{2}\sigma_z$$







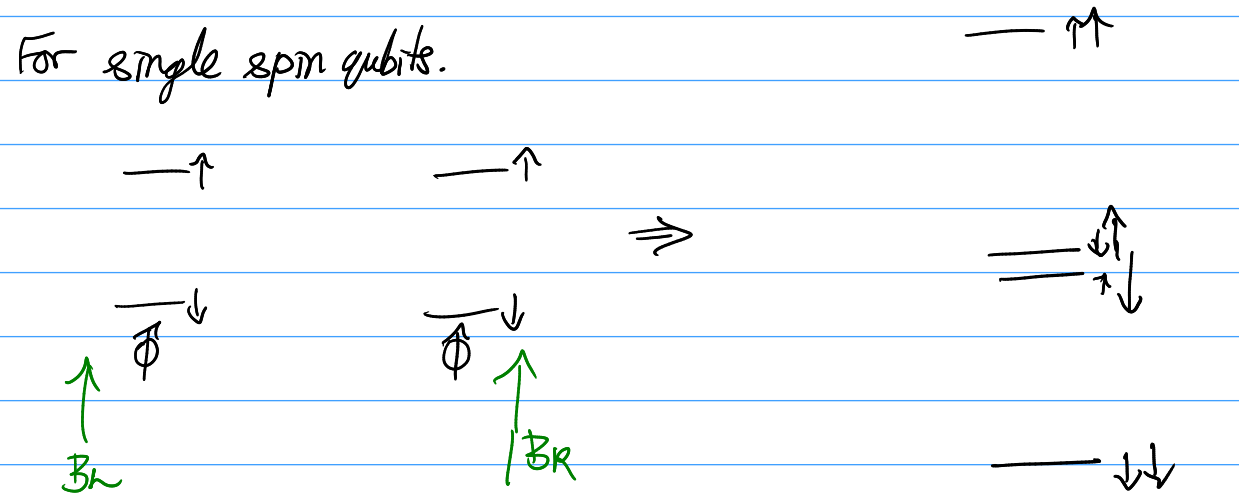
$$H_{rot} = \frac{B_z - \hbar\omega}{2} + \cos(\omega t + \varphi)\sigma_x + \sin(\omega t + \varphi)\sigma_y$$

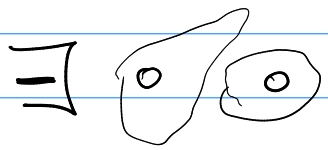
That is, for ESR or even transmons/ion traps/atomic systems etc.:

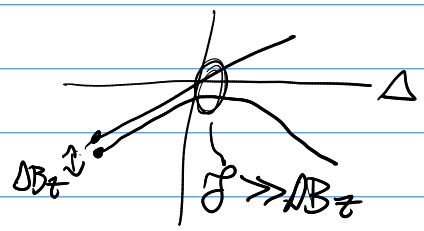
- Drive RF at $\hbar\omega = B_z$ for x-rotations. ($H = \frac{E_z - \hbar\omega}{2}\sigma_z + \hbar g\sqrt{N+1}\,\sigma_x$)
- Shift phase of remaining gates for z-rotations
- Shift phase of current gate for y-rotations. (e.g. 90°)
- Change drive amplitude for faster/slower rotations.

2- Qubit Gates

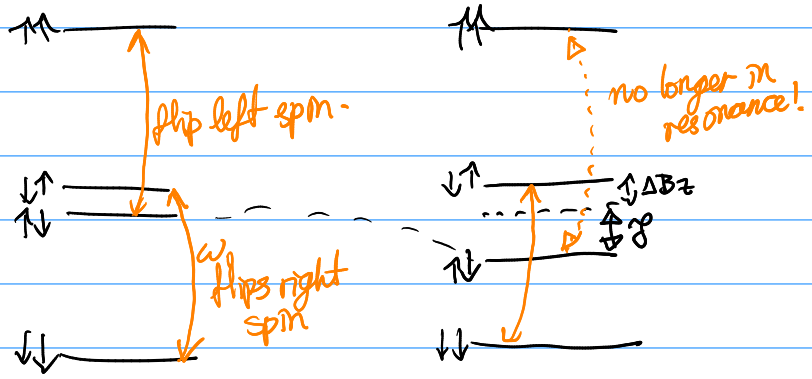
For single spin qubits.



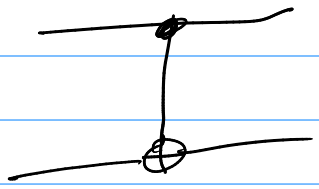




$\Delta \gg J$ or $J \gg 0$



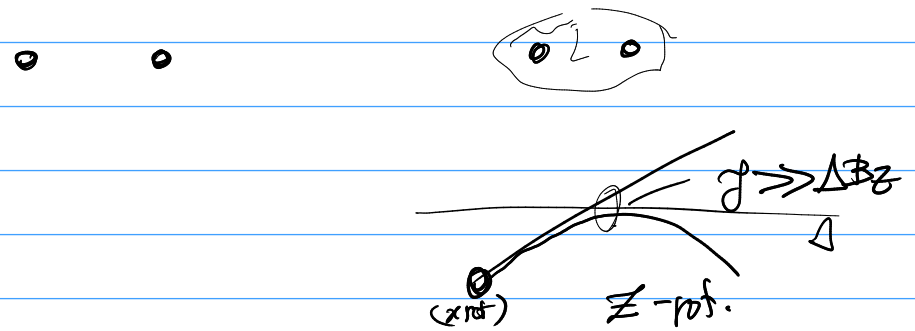
or flip if left spin is down.



$$H_{\substack{1st\ spin \\ down}} = B_{ac}\sigma_x$$

$$H_{\substack{1st\ spin \\ \uparrow}} = 0$$

For S-T. qubits.



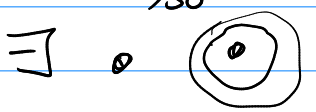
Idle- $H = J\sigma_z$

To



$H = J\sigma_z$

So



$H = (J + \Delta J)\sigma_z$

$\Rightarrow$ e^- / $\hat{z}$-rot. is $\sim \Delta J$ faster if 1st qubit is in So state.

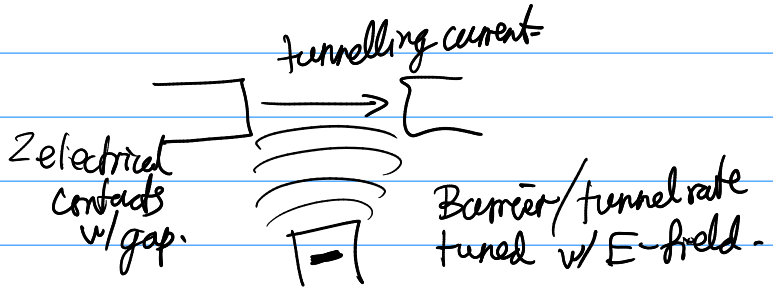
(electrostatic repulsion $\Rightarrow$ more Δ)

If wait ½ period of ΔJ (controlled on 1st qubit being So)

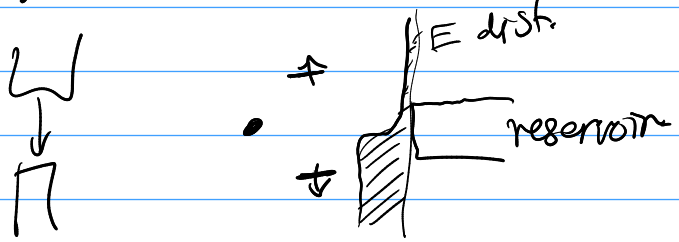


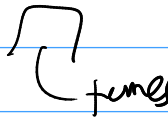
Readout

Charge sensor

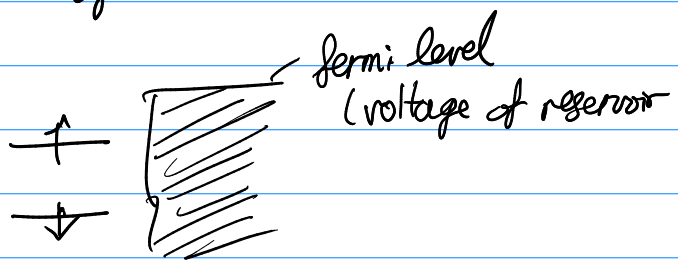


Single spins.

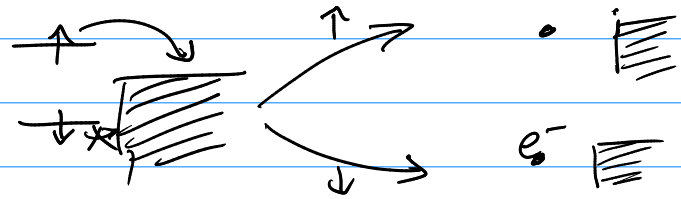




Idle.



Readout

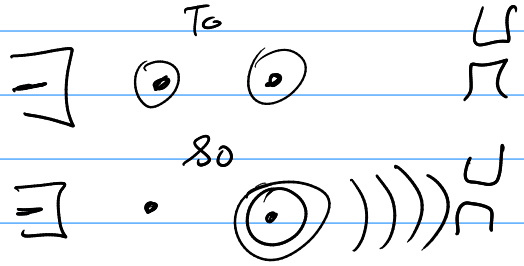


Charge sensor has diff tunnelling current.
Complete destructive measurement.

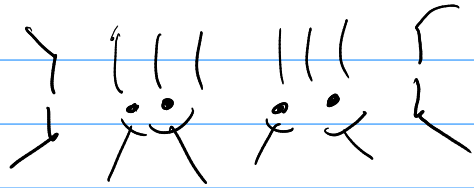
'Elzerman or single spin readout'

So / To.

Pauli Blockade readout

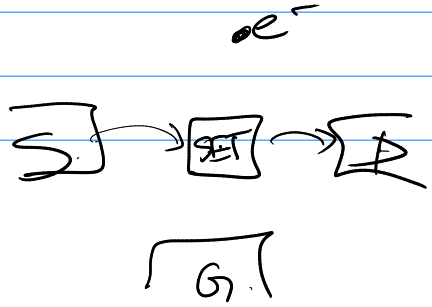


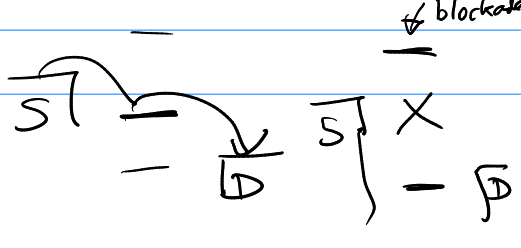
Shulman 2012



hard to scale up.

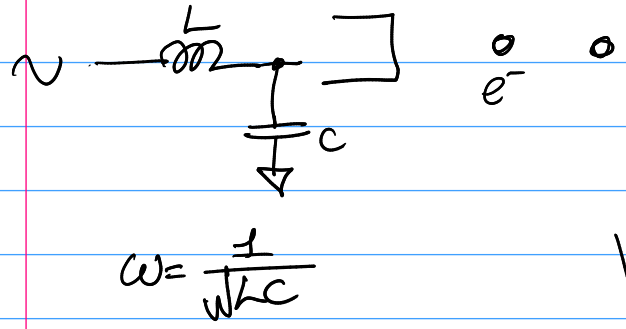
(SET.)

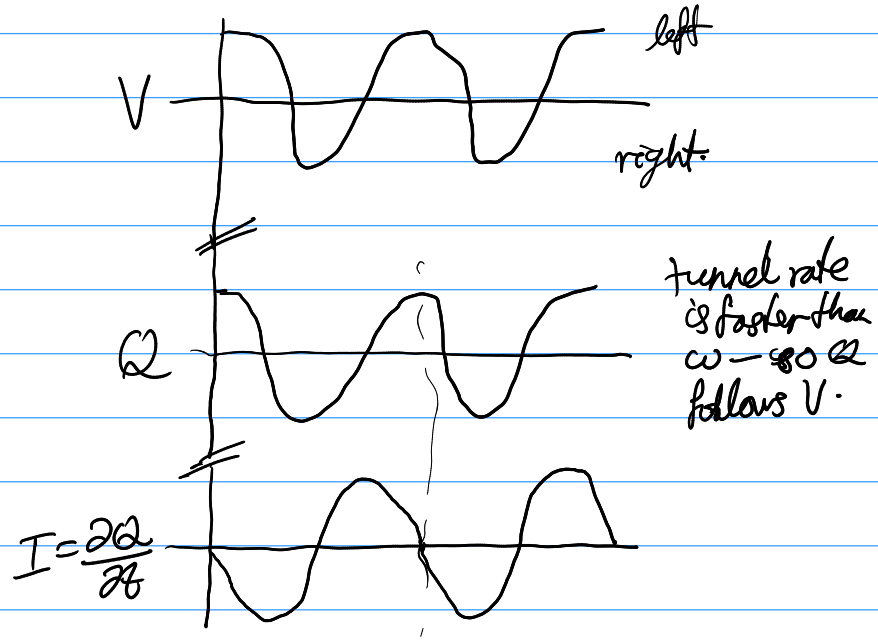




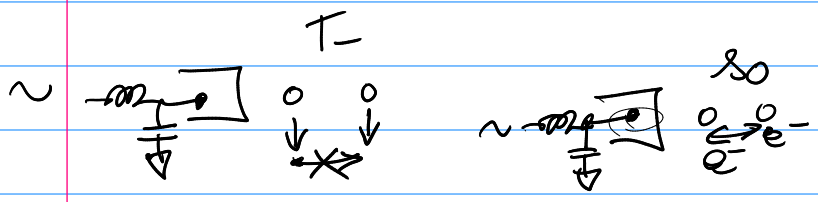
RF gate-based readout (S_0/T_0)

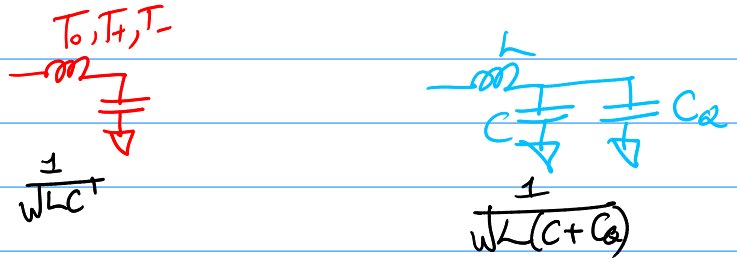


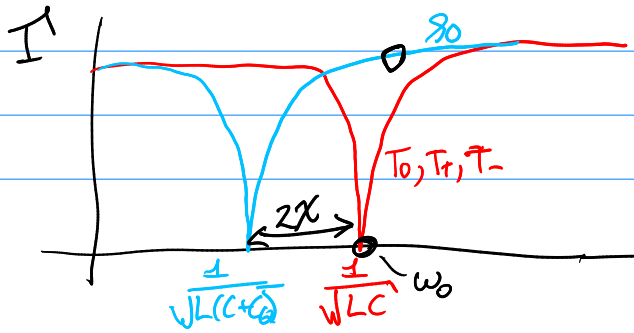




I is 90° out of phase w/ V
$\Rightarrow$ a.c. capacitance:



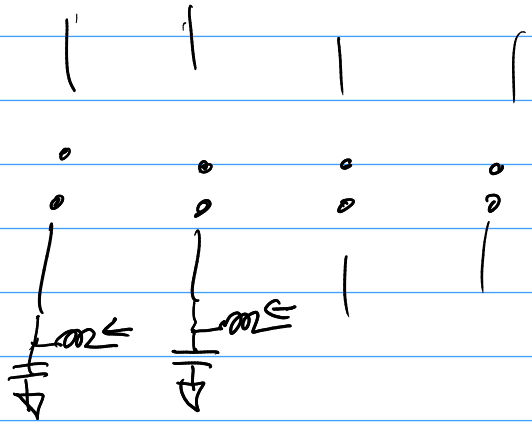




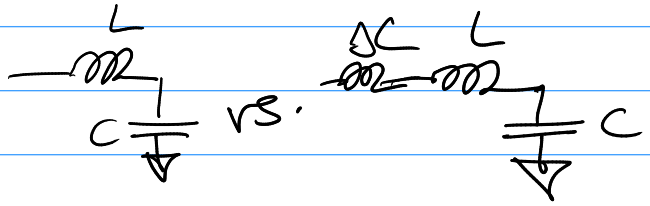
Signal contrast in T_+ depends on 2X (large is good)

$\Rightarrow$ Also want high Q-factor. (i.e. narrow linewidth)





Pakkiam Nano Lett.



(Transmon Qubits.)

$\frac{1}{\sqrt{LC}}$ $\qquad$ $\frac{1}{\sqrt{(L+\Delta L)C}}$